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# Artem Makoian

**Programming:** Python 3, C++  
**Languages:** English (C1), Russian (native)  
github.com/MakArtKar  
scholar.google.com  
leetcode.com/MakArtKar

## SKILLS

**Libraries** PyTorch, JAX, Transformers, Hydra, DVC, WandB, OpenCV, Sklearn, Numpy, Pandas, ONNX, Seaborn  
**Tools** SLURM (DDP, FSDP), SQL, Linux, Bash, Git,  $\text{\LaTeX}$ , Markdown

## TECHNICAL EXPERIENCE

 **JetBrains** Feb 2025 — Present  
ML Researcher / JetBrains Research *with D. Vetrov* Python 3, PyTorch, JAX, transformers, Hydra, Lightning, SLURM

→Masked Diffusion LMs

- Inference-time scaling, diffusion drafters for speculative decoding, curriculum learning (Dream-7B, Qwen3-0.6B MDLM/BD3LM)
- **G-Star**: ICML 2026 main track; multi-node training; PyTorch, JAX

→Latent Diffusion Drafters for speculative decoding: lossless acceleration of LLM inference

- Distillation (flow maps, consistency) and latent space compression for real-time acceleration

 **Yandex** Jun 2023 — Sep 2024  
Middle ML Engineer / Geo services Python 3, LangChain, PyTorch, Hydra, OpenCV, YT, Shapely, YQL, Nirvana

→Road segmentation from aerial imagery with IoU 0.93

- Roads post-processing algorithms - reduced number of artifacts from 10% → 0.02% and improved visual quality.
- Safety islands segmentation and vectorization with IoU 0.37

→News classification to create map updates hypotheses. Precision: 0.9, recall: 0.85.

- Developing and maintenance of the news pipeline: site parsing (with LangChain AI agents), structuring data and classifying.

 **Sber Devices** Jul 2022 — Jun 2023  
Middle ML Engineer / Computer Vision Platform Python 3, C++, PyTorch, Hydra, DVC, OpenCV, ONNX, torch\_pruning

→Background removal for video conferencing with 0.98 Dice score

- Improved human segmentation quality working on segmentation pipeline, pruned model getting 1.5x speed acceleration
- Improved visual quality with post-processing algorithms: aligning the lighting and smoothing out the border.

→Trained a 700 KB model for payment card border segmentation and developed C++ inference pipeline in multiplatform mobile SDK

 **AIRI (Artificial Intelligence Research Institute)** Jul 2022 — Oct 2022  
ML Research Intern / GAN adaptation Python 3, PyTorch, Hydra, WandB

- **Few-Shot Image Generation** implementation for few-shot GAN adaptation to a new domain from ~10 images

 **Yandex** Jul 2021 — Oct 2021  
ML Researcher / VR Team *with V. Lempitsky* Python 3, PyTorch, Tensorboard

→Trained model that predicts human's alpha mask to improve the quality of a VR scene

- Prepared datasets with human alpha masks, read and compared papers on an alpha matting problem
- Trained an improved **FBA Matting** net without using trimaps

 **Samsung R&D Department** Nov 2020 — Jun 2021  
ML Research Intern / Mobile Authentication Python 3, PyTorch, Bash, Numpy, Pandas, Sklearn, Tslearn

→Worked on a pipeline for mobile phone authentication by user's motion patterns with  $82.2\% \pm 9\%$  accuracy - [paper](#).

- Collected, preprocessed and handled data from devices. Used ML and DL models for authentication, got 70% accuracy baseline

## EDUCATION

 **PhD in Computer Science (1st year)**, *Constructor University, Bremen, Germany* Sep 2026 — Present

- Teaching Assistant (seminars): Deep Bayesian Models (Spring 2026), Bayesian Methods (Fall 2026)
- Co-supervising 3 MSc theses: Loop Transformers, Gaussian Diffusion LMs, Jacobi Decoding

 **Master in Computer Science**, *Constructor University, Bremen, Germany* Sep 2024 — May 2026

 **Bachelor in Computer Science**, *Neapolis University of Paphos, Cyprus*, GPA 86/100 Jan 2023 — Jun 2024

 **Bachelor in Computer Science**, *HSE University, Moscow, Russia*, GPA 8.7/10 Sep 2019 — Jun 2024

 **Master's Equivalent in Data Science**, *Yandex School of Data Analysis, Russia* Sep 2020 — May 2022

*Selected courses:* Machine Learning, Deep Learning, Generative Models, Bayesian Methods, Optimization Methods, Large-Scale DL, NLP, Computer Vision, Deep Vision, 3D Computer Vision, Reinforcement Learning, Probability Theory, Statistics, Linear Algebra, Matrix Computations, Algorithms and DS